Examiner only

5. Electromagnetic (em) waves are used extensively in a wide variety of applications. The table below gives information about different regions of the electromagnetic (em) spectrum. It shows ranges for wavelengths and frequencies.

Region	Wavelength (m)	Frequency (Hz)
radio	$> 1 \times 10^{-1}$	....
microwave	1×10^{-3} to 1×10^{-1}	3×10^{9} to 3×10^{11}
infra-red	....	3×10^{11} to 4×10^{14}
visible	....	4×10^{14} to 7.5×10^{14}
ultraviolet	1×10^{-8} to 4×10^{-7}	7.5×10^{14} to 3×10^{16}
X-ray	1×10^{-11} to 1×10^{-8}	3×10^{16} to 3×10^{19}
gamma ray	$< 1 \times 10^{-11}$	....

(a) **Complete the table** above using the ranges below. [3]

(7.5×10^{-7} to 1×10^{-3}) **($< 3 \times 10^{9}$)** **($> 3 \times 10^{19}$)** **(4×10^{-7} to 7.5×10^{-7})**

(b) All em waves transfer energy. The shortest waves are the most ionising. The table below shows energy values in an **incorrect order**. Arrange the energy values in the **correct order.** [1]

Region	Energy (J) in **incorrect order**	Energy (J) in **correct order**
radio	$> 2 \times 10^{-14}$	....
microwave	3×10^{-19} to 5×10^{-19}	....
visible	2×10^{-17} to 2×10^{-14}	....
X-ray	2×10^{-24} to 2×10^{-22}	....
gamma ray	$< 2 \times 10^{-24}$	....

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